

A project report on
CUSTOMER SALES ANALYSIS



THE DEVELOPER ARENA

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Project Overview

This project focuses on analyzing customer sales data to understand purchasing patterns, regional sales performance, and product demand. The analysis helps businesses gain insights into their sales performance and customer behavior.

The project uses Python-based data analysis techniques to process datasets and generate meaningful visualizations and insights.

The main goal is to demonstrate how data analysis can help organizations make informed business decisions.

Setup Instructions

Follow these steps to run the project:

- Open Google Colab or any Python notebook environment.
- Upload the datasets:

Sales_data.csv

Customer_churn.csv

- Install the required libraries:
Pip install pandas numpy matplotlib seaborn
Open the notebook file customer_analysis.ipynb.
- Run all cells to generate analysis results and visualizations

Code Structure

The project is organized in a clear and structured workflow:

Importing required libraries

Loading datasets

Data cleaning and preprocessing

Merging datasets

Performing aggregations

Creating pivot tables

Generating visualizations

Extracting insights from the results

This structure ensures readability and makes the analysis process easy to understand

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Loading datasets...")

sales = pd.read_csv("sales_data (3).csv")
churn = pd.read_csv("customer_churn.csv")

# -----
# Preview datasets
# -----

print("\nSales Data")
print(sales.head())

print("\nChurn Data")
print(churn.head())

# -----
# Data Cleaning
# -----

sales.dropna(inplace=True)
churn.dropna(inplace=True)

# -----
# Merge datasets
# -----

data = pd.merge(sales, churn, left_on="Customer_ID", right_on="CustomerID", how="left")

print("\nMerged Data Shape:", data.shape)

# -----
# Aggregation 1 - Total Revenue
# -----

total_revenue = data["Total_Sales"].sum()

# -----
# Aggregation 2 - Sales by Region
# -----

region_sales = data.groupby("Region")["Total_Sales"].sum()

# -----
# Aggregation 3 - Product Sales
# -----

product_sales = data.groupby("Product")["Total_Sales"].sum()

# -----
# Average Order Value
# -----

avg_order = data["Total_Sales"].mean()

# -----
# Top Customers
# -----

top_customers = data.groupby("Customer_ID")["Total_Sales"].sum().sort_values(ascending=False)

# -----
# Pivot Table
# -----

pivot = pd.pivot_table(
    data,
    values="Total_Sales",
    index="Region",
    columns="Product",
    aggfunc="sum"
)

```

```

# Churn Analysis
# -----

churn_analysis = churn["Churn"].value_counts()

# -----
# Print Results
# -----

print("\n===== CUSTOMER SALES ANALYSIS REPORT =====")

print("\nTotal Revenue:", total_revenue)

print("\nAverage Order Value:", avg_order)

print("\nSales by Region:")
print(region_sales)

print("\nTop Customers:")
print(top_customers)

print("\nProduct Sales:")
print(product_sales)

print("\nPivot Table:")
print(pivot)

print("\nCustomer Churn Distribution:")
print(churn_analysis)

# -----
# Visualizations
# -----

plt.figure()
region_sales.plot(kind="bar")
plt.title("Sales by Region")
plt.ylabel("Revenue")
plt.show()

plt.figure()
product_sales.plot(kind="bar")
plt.title("Product Sales")
plt.ylabel="Revenue"
plt.show()

plt.figure()
product_sales.plot(kind="pie", autopct="%1.1f%%")
plt.title("Product Sales Distribution")
plt.show()

plt.figure()
sns.heatmap(pivot, annot=True, fmt=".0f")
plt.title("Sales Heatmap")
plt.show()

plt.figure()
churn_analysis.plot(kind="bar")
plt.title("Customer Churn")
plt.show()

print("\nAnalysis Completed Successfully")

```

Visual Documentation

The project includes visualizations to better understand the data:

Sales by Region – bar chart showing revenue distribution across regions.

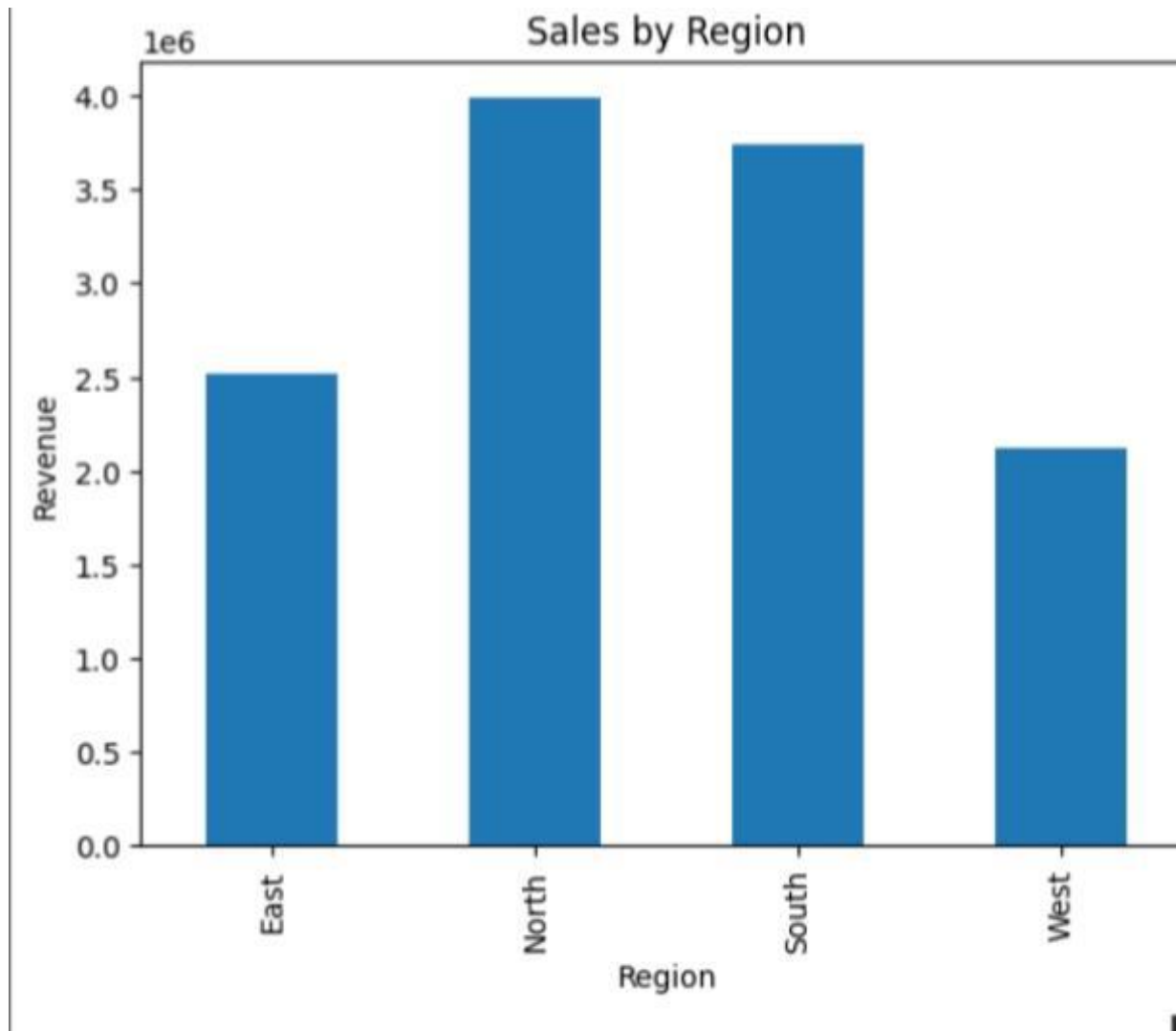
Product Sales – bar chart representing product performance.

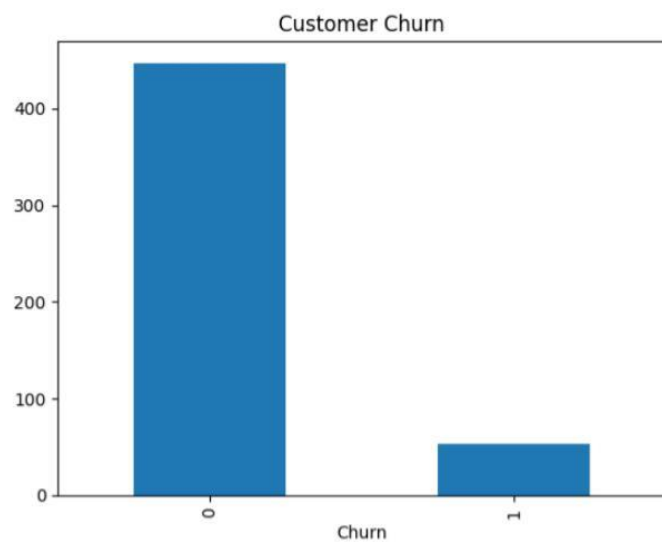
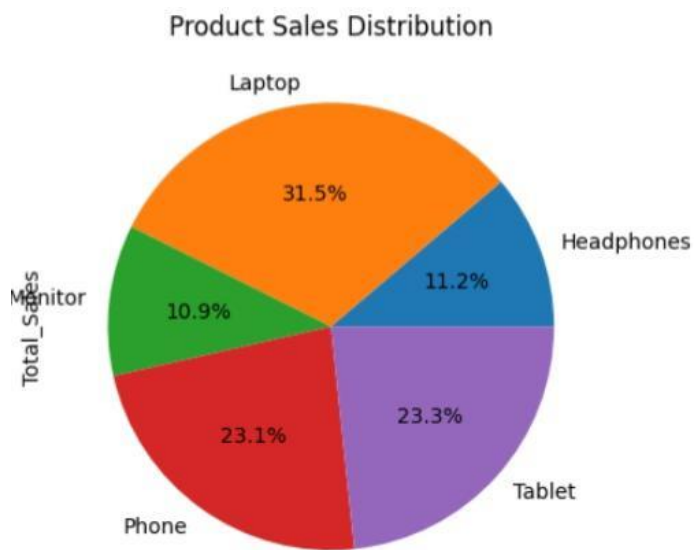
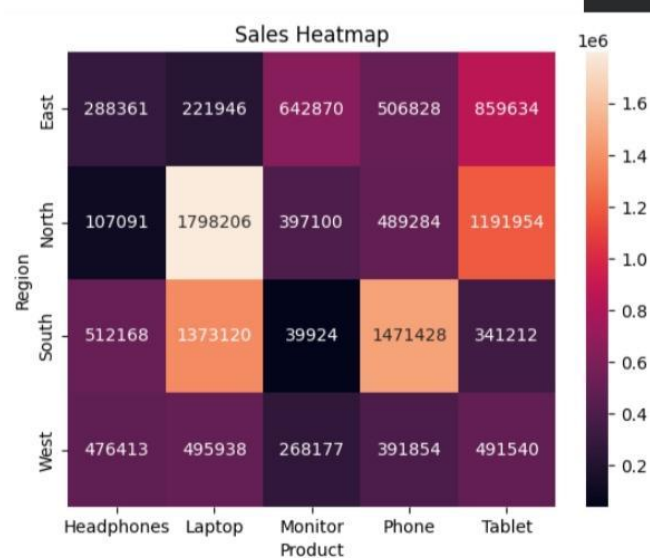
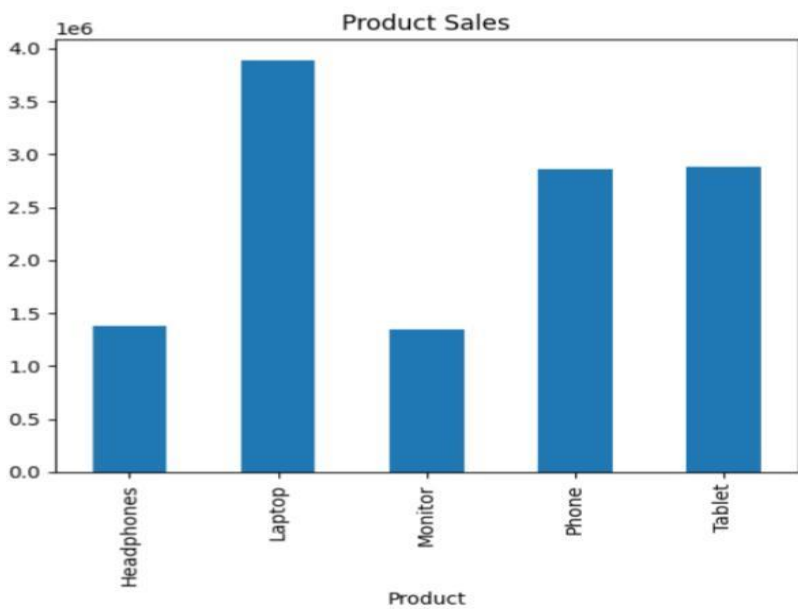
Product Distribution – pie chart showing percentage contribution of each product.

Sales Heatmap – heatmap visualization using pivot tables.

Customer Churn Distribution – chart showing churned vs retained customers.

These visualizations help interpret the data more effectively





Technical Details

The project uses the following technologies and techniques:

Programming Language

Python

Libraries Used

Pandas for data manipulation

NumPy for numerical operations

Matplotlib for visualizations

Seaborn for advanced data visualization

Techniques Applied

Data cleaning

Dataset merging

Aggregation operations

Pivot tables

Data visualization

The main data structure used in this project is the Pandas DataFrame, which allows efficient data analysis

Testing Evidence

Testing was performed by executing the notebook with the provided datasets.

Validation steps included:

Verifying dataset structure and columns

Checking merged dataset correctness

Confirming aggregation results

Ensuring charts and visualizations are generated correctly

These tests ensure the analysis results are accurate and reliable.

```
Loading datasets...

Sales Data
  Date      Product  Quantity  Price  Customer_ID  Region  Total_Sales
0  2024-01-01   Phone        7  37300   CUST001     East    261100
1  2024-01-02 Headphones        4  15406   CUST002    North    61624
2  2024-01-03   Phone        2  21746   CUST003     West    43492
3  2024-01-04 Headphones        1  30895   CUST004     East    30895
4  2024-01-05   Laptop        8  39835   CUST005    North    318680

Churn Data
  CustomerID  Tenure  MonthlyCharges  TotalCharges  Contract \
0  C00001        6         64         1540      One year
1  C00002       21        113        1753  Month-to-month
2  C00003       27         31        1455      Two year
3  C00004       53         29        7150  Month-to-month
4  C00005       16        185        1023      One year

PaymentMethod  PaperlessBilling  SeniorCitizen  Churn
0  Credit Card                No                1        0
1  Electronic Check            Yes                1        0
2  Credit Card                No                1        0
3  Electronic Check            No                1        0
4  Electronic Check            No                1        0

Merged Data Shape: (100, 16)

===== CUSTOMER SALES ANALYSIS REPORT =====

Total Revenue: 12365048

Average Order Value: 123650.48

Sales by Region:
Region
East    2519639
North   3983635
South   3737852
West    2123922
Name: Total_Sales, dtype: int64

Top Customers:
Customer_ID
CUST016    373932
CUST007    363870
CUST083    350888
CUST073    349510
CUST020    333992
Name: Total_Sales, dtype: int64

Product Sales:
Product
Headphones  1384033
Laptop      3889210
Monitor     1348071
Phone       2859394
Tablet      2884340
Name: Total_Sales, dtype: int64
```

Conclusion

This project demonstrates how data analysis techniques can be used to understand customer behavior and sales performance.

By analyzing sales and churn data, businesses can identify trends, evaluate product demand, and improve customer retention strategies. The visualizations and insights generated in this project highlight the importance of data-driven decision-making in modern business environments